



Armada Kalamunda Group

Safe Prescribing, Storage, Supply, Preparation and Administration of Potassium Guideline

1. Purpose

This guideline defines processes for the safe prescribing, storage, supply, preparation, and administration of potassium at Armada Kalamunda Group (AKG); and outlines the potassium products which are available for use at AKG.

2. Applicability

This guideline applies to medical, nursing, pharmacy, and all services within AKG.

3. Prescribing

Oral or enteral potassium supplementation should be used where possible.

- Potassium chloride (KCl) is to be used when intravenous (IV) potassium is ordered unless otherwise specified.
- Orders must be prescribed on the Intravenous Fluid Order Chart and written in full. EXCEPTION: AKMR 176.1.1 Standing Order Potassium Chloride (KCl) Replacement Protocol form in use (Intensive Care Unit only)
- All IV potassium orders must be prescribed in millimoles (mmol) – NOT grams, percentages, or milligrams/litre.
- Prescriptions for IV potassium must also include duration and/or rate of infusion i.e., mL/hour. The rates should not exceed the maximum rates specified in this guideline.
- Premixed potassium infusion bags are to be used for all IV potassium administration. Under exceptional circumstances, non-standard concentrations of potassium may be permitted when clinically desirable and with the explicit agreement of the consultant in charge. The consultant must either:
 - Prescribe the non-standard order, or
 - The non-standard order must be annotated with the consultant's signature.
- Where possible, the ward or dispensary pharmacist must be notified if a non-premixed solution is prescribed.
- To replace potassium without giving large amounts of fluid, repeated doses of 10mmol in 100mL may be given e.g., to give 40mmol, give four 10mmol KCl in 100mL bags. This allows the flexibility of checking plasma potassium at 20mmol and reviewing the need for further potassium supplementation.
- If pain occurs during administration of intravenous potassium, consider reducing the rate of infusion or prescribing a more dilute preparation.
- For appropriate intravenous potassium replacement in paediatric patients, refer to Perth Children's Hospital Guidelines [Potassium Chloride - Paediatric](#).

Before referencing this guideline document please ensure that you have the latest version from the [Policy Library](#) information hub page.

- For a list of potassium products available at AKG, refer to Appendix 1.

3.1 Administration and Monitoring of Intravenous Potassium

- All infusions containing potassium must be administered via an infusion pump
- The **maximum rate** of potassium chloride administration via **peripheral** IV lines is 10 mmol/hour. The **maximum concentration** of potassium chloride for administration via **peripheral** IV lines is 40 mmol/L (except for pre-mixed 10 mmol/100mL isotonic)
- For **PAEDIATRIC PATIENTS**, the maximum rate is 0.2 mmol/kg/hour.
- More concentrated solutions and faster rates of administration are to be infused via central line (CVC).
- **Continuous ECG monitoring is required when the rate is faster than 10 mmol/hour.** Rates in excess of 20 mmol/hr are potentially hazardous.
- Administration of a regimen that differs from this guideline must be approved in advance by the treating physician/specialist and documented on the intravenous fluid order chart.
- In an emergency, the infusion may be started through a peripheral line until a central venous line is inserted.
- Potassium dihydrogen phosphate is infused at a slower rate to reduce the risk of hypocalcaemia e.g., 10 mmol diluted in compatible fluid and infused over 2-12 hours.
- Monitor serum or ABG Potassium levels every 2 hours during continuous potassium replacement.
- All patients being treated with intravenous potassium should have at least daily measurements of plasma potassium until levels are shown to be within the normal range for plasma potassium and stable.

3.2 Storage and Supply

- Potassium chloride ampoules and other forms of concentrated injectable potassium will NOT be stored on general wards.
- The storage of high concentration potassium products is restricted to the Pharmacy Department and ICU.
- Potassium chloride ampoules must be stored
 - In a locked cupboard, except where storage limitations exist and arrangements for alternative secure storage are made with the Pharmacy Department.
 - In clearly marked, red containers and isolated from all other ampoules
- If a non-standard concentration of IV potassium is required on general ward areas, Pharmacy will supply only the required number of ampoules for that infusion.
- Any unused ampoules of IV potassium must be returned to the Pharmacy as soon as possible.
- Do NOT place or store potassium ampoules on resuscitation trolleys.

3.4 Obtaining Ampoules After Hours

- The treating team may be contacted to discuss whether a pre-mixed preparation is suitable.
- The ICU is not permitted to dispense Potassium ampoules to other areas of the hospital unless explicitly advised by a Clinical Pharmacist.

- After-hours access to potassium chloride ampoules and other forms of concentrated injectable potassium is restricted to the After-Hours Nurse Manager.
- Any unused IV potassium ampoules taken after hours must be returned to the Pharmacy Department as soon as possible.

3.5 Preparation of Non-Standard Potassium Solutions

Non-standard potassium chloride infusions should ONLY be prepared IF the standard pre-mixed solutions CANNOT be used and the consultant in charge has given explicit agreement. The name of the consultant providing the approval must be documented on the order. This is the responsibility of the prescriber.

- Potassium ampoules MUST be diluted before administration to a patient.
- Where non-standard potassium solutions are prepared, the solution MUST be well-mixed.
- Fully invert the bag AT LEAST ten times and agitate and/or knead to ensure adequate mixing.
- Unshaken bags are prone to layering of added concentrate and are extremely hazardous.
- NEVER add additional potassium to a pre-mixed solution.
- NEVER add potassium directly to a running/hanging IV bag.
- Two authorised clinical staff, as per usual IV preparation protocols, must check the preparation, dose, dilution, labelling, route and rate, and accordance with the 6 rights of medication administration.

3.6 Nursing Management

- Ensure patent intravenous access.
- Monitor PIVAS (peripheral intravenous assessment score) three times a day and as required.
- If extravasation occurs, stop the infusion immediately, inform the treating doctor and prepare for the insertion of a new intravenous cannula.
- If pain occurs during administration via peripheral line, inform the treating doctor. The rate of infusion may be reduced or a more dilute preparation

4. Legislative context

The following documents are required to give effect to this policy [i.e. the documents included are mandatory:

- Medicines and Poisons Act 2014
- Health Department of Western Australia (HDWA) High Risk Medication Policy MP 0131/20
- HDWA Medication Review Policy MP 0104/19
- HDWA Clinical Handover Policy MP 0095/18
- HDWA Clinical Incident Management Policy MP 0122/19

- HDWA Credentialing and Defining Scope of Clinical Practice Policy MP 0084/18
- HDWA State-wide Medicines Formulary Policy MP 0077/18
- East Metropolitan Health Service (EMHS) Patient Identification and Procedure Matching EMHS: 21

5. Supporting information

The following documents support the implementation of this policy:

- Medication Management AKG-PRO-0255-18
- Adult Diabetic Ketoacidosis GEN-GUI-0012-15

6. Compliance, monitoring and evaluation

Compliance against this policy will be evaluated with review of potassium prescriptions to ensure compliance; review storage and access of IV potassium chloride preparations in the pharmacy and on wards to reduce the likelihood of substitution errors.

7. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition):

- Standard 4 – Action 4.3; 4.4; 4.14; 4.15

8. References

1. Hypokalaemia [amended June 2015]. In: eTG complete [Internet]. Melbourne: Therapeutic Guidelines Limited; 2018 July.
2. Potassium Chloride Accessed from aidh.hcn.com.au. (2022). Australian Injectable Drugs Handbook, 7th Edition; The Society of Hospital Pharmacists of Australia [online]
3. Potassium Dihydrogen Phosphate Accessed from aidh.hcn.com.au. (2022). Australian Injectable Drugs Handbook, 7th Edition; The Society of Hospital Pharmacists of Australia [online]
4. Potassium Supplementation Clinical Guideline, Royal Perth Bentley Group, 2021

9. Glossary

The following definitions are relevant to this policy.

Term	Definition
KCl	Chemical abbreviation for potassium chloride. Chemical abbreviations are not to be used when prescribing potassium supplementation
K+	Potassium elemental. Chemical abbreviations are not to be used when prescribing potassium supplementation
Nonstandard	An intravenous potassium order which deviates from the standard concentrations of the pre-mixed parenteral preparations available. Permitted only under the explicit

Term	Definition
	direction of the consultant in charge and in exceptional circumstances.

10. Document contact

Enquiries relating to this guideline may be directed to:

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11. Review

This guideline will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

Version	Effective from	Effective to	Amendment(s)
V1	07/09/2023	07/09/2026	New document

12. Authorisation

This guideline has been authorised and issued by the [enter position title of the authoriser].

Authorised by	A/Deputy Director Clinical Services
Authorisation date	06/09/2023
Published date	07/09/2023

Appendix 1: Potassium supplementation available at AKG

ORAL Potassium Products available at Armadale Kalamunda Group	Quantity of potassium (K+) per dosage unit (comments)
Potassium chloride SR (slow release) 600 mg	8 mmol K+ per tablet
	For oral use only; Not to be crushed or chewed
Potassium effervescent salts; (e.g. Chlorvescent®)	14 mmol K+ per tablet
	For oral, nasogastric or PEG; Disperse each tablet in 120mL of water prior to administration
Potassium chloride elixir	20 mmol K+ in 15 mL
	For oral, nasogastric or PEG; Dilute with 50mL-100mL fluid prior to administration

PARENTERAL Preparations available at Armadale Kalamunda Group	Quantity of potassium (K+) per dosage unit	BAXTER CODE
1. Pre-mixed Intravenous Bags		
Potassium chloride 10mmol in sodium chloride 0.29% 100 mL (isotonic)	10mmol K+ in 100 mL	AHB6008
Potassium chloride 20mmol in sodium chloride 0.9% 1000 mL	20mmol K+ in 1000 mL	AHB1764
Potassium chloride 20mmol in sodium chloride 0.18% glucose 4% 1000 mL	20mmol K+ in 1000 mL	AHB1704
Potassium chloride 20mmol in glucose 5% 1000 mL	20 mmol K+ in 1000 mL	AHB1134
Potassium chloride 40mmol in sodium chloride 0.9% 1000 mL	40 mmol K+ in 1000 mL	AHB6034
Potassium chloride 20mmol in sodium chloride 0.9% glucose 5% 1000 mL	20 mmol K+ in 1000 mL	AHB6066

Potassium chloride 30mmol in Hartmann's solution (modified) 1000 mL [pharmacy requires 72 hours notice]	30 mmol K ⁺ in 1000 mL	AHB2954
2. Potassium (concentrated) Ampoules		
RESTRICTED ACCESS – CONSULT PHARMACY: AMPOULES (CONCENTRATED) MUST BE DILUTED BEFORE USE		
Potassium acetate (concentrated)	25 mmol K ⁺ in 5 mL	
Potassium dihydrogen phosphate (concentrated)	10 mmol K ⁺ in 10 mL	
Potassium chloride (concentrated)	10 mmol K ⁺ in 10 mL	